

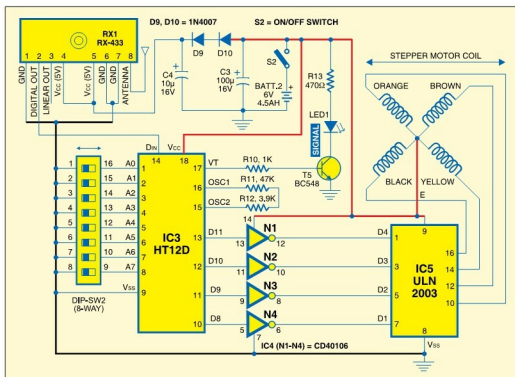


Fig. 7: Receiver-cum-decoder circuit for wireless stepper motor control

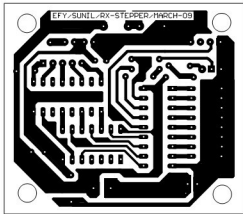


Fig. 8: Actual-size, single-side PCB for the receiver-cum-decoder circuit

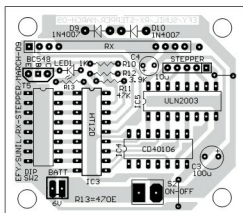


Fig. 9: Component layout for the PCB shown in Fig. 8

Once the frequency of the pair is aligned, then on ground of any data pin of the encoder, LED1 of the decoder should glow. You can also check the transfer of data on pins AD8 through AD11, which is latched to D8 through D11 pins of the decoder once TE pin is momentarily taken low by grounding it through diodes D1 through D4.

RF transmitter and receiver. RF transmitter and receiver modules from Alpuz India, Mumbai, have been used for RF remote control.

The RF transmitter TX-433 is AM/ASK type. Its features include:

1. 5V-12V single-supply operation
2. On-off keying (OOK)/amplitude shift keying (ASK) data format
3. Up to 9.6kbps data rate
4. +9dBm output power (about 200m range)
5. SAW-based architecture
6. A 45cm wire is adequate for the antenna

The output power and current drain for Vcc of 5V and 12V are

shown in Table I.

The pin configuration of the transmitter module is shown in Fig. 2.

The RF receiver RX-433 is a 433MHz module. Its pin configuration is shown in Fig. 3, while the technical specifications are given in Table II.

Transmitter. Fig. 4 shows the circuit of the transmitter for wireless stepper motor control. The receiver address to be transmitted can be set with the help of 8-way DIP switch SW1. When any of the switch contacts is open the respective pin will be at logic 1, and when any of the switch contacts is closed the respective pin will be

at logic 0. The data pins are pulled high via resistors R2 through R5.

When pin 2 of the parallel port goes high, transistor T1 is driven into saturation and relay RL1 energises. Pin 10 (AD8) goes low through relay RL1 contacts and a '0' is sent at that data position, while other data pins represent logic-1 state. The logic circuitry at the receiver-decoder end decodes the data appropriately for controlling the stepper motor.

An actual-size, single-side PCB for the transmitter circuit (Fig. 4) is shown in Fig. 5 and its component layout in Fig. 6.

Receiver and decoder. Fig. 7 shows the circuit of the receiver-cum-decoder for wireless stepper motor control. Assuming that identical address is selected on the encoder and decoder, when any of the data pins of the PC's parallel port on the transmitter side is low, the corresponding data pin of the decoder will go low. The data outputs (D8 through D11) of HT12D are fed to inverters N1 through N4, which, in turn, are connected to driver ULN2003. The low output of ULN2003 drives the stepper motor. When any